Name..... **Signature**.....

Candidates Personal Number.....

545/1

CHEMISTRY.

Paper 1.

Aug /2025.

2 hours

CHEMISTRY DEPARTMENT

New Lower Secondary Chemistry Examination

Senior 3

INSTRUCTIONS:

This paper consists of **two** *sections* **A** *and* **B**. *It has* **five** *examination items. Section* **A** *has* **one compulsory** items.

Section **B** *has* **two parts, I** and **II**. *Answer* **one** item from each part. Answer four items in all.

Answers to section **A** **must** *be written in the spaces provided while those of section* **B** **must** *be written on the provided answer sheets.*

Any additional item(s) answered will **not** *be scored*

Where necessary use C = 12, H = 1, O = 16.

SECTION A.

Answer **all** *items in this section*

Item 1.

A Chemical investor intends to set up a small scale production facility using glucose obtained from locally sourced materials to produce useful organic product as equation below. They need to conduct preliminary experiment and need to know the mass of organic that can be produced from 0.5kg of glucose. The results obtained can enable them to scale up to target production. You have been invited for guidance also inform them environmental responsibility of this reaction

$$C_6H_{12}O_6\ (aq) \xrightarrow{\quad} 2C_2H_5OH(l) + 2CO_2(g)$$

Task

As a learner of chemistry,

(a) explain the category of organic product.

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(b) Suggest the suitability of organic product to its applications in every day life.

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(c) Calculate the mass of organic product produced in above reaction

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(d) explain the environmental impact associated with the gaseous product.

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Item 2.

Two students, Mary and Jane live in a rural area of a rocky terrain. Their household water source is a stream that flows directly from a spring originating within the nearby granite rock formation. One afternoon, each washed identical white cotton shirts using different available cleaning products with their steam water. Later , their mother observed that Jane used significantly more product than Mary to achieve what appeared to be similar level of cleanliness.

Task. As a learner of chemistry,

(a) Categorize the products used by two students.

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(b) Explain the cleaning action of the product used by any of the two students

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(c) What challenges are associated with the product Jane used.

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(d) Evaluate the products used by the two students.

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SECTION B

Part I

Attempt one item from this part

2. Due to high demand of soapy detergents on market, the government has cleared a local investor to establish a mega soapy detergent production industry with minimal impact. However, the science club members in your school would like to know how the process of production will be carried out.

Task. As a chemistry student, make a write up you will use during the presentation.

3. Due to increased demand of oxygen gas in many health centers, the government has cleared an investor to set an oxygen production facility in Makindye . However the locals are concerned about it's impact resulting from the process of production. You have been invited to address the locals in one of the sensitization workshops conducted at the sub county hall

Task. Write a write up presentation you will use during the presentation.

Part II.(*Attempt one item from this part.*)

4. The district officials of Kisoro have raised concern about the destruction of the rocky areas due to increasing population in the region. This has prompted the officials to organize sensitization workshops in the region about environmental protection and sustainability.

Task. As a chemistry prepare a short presentation you will deliver in the workshop.

5 The National Environmental Management Authority (NEMA) has raised concern about the impact of the industrial set up on the nearby water sources. This has prompted the government to conduct regional radio talk shows to raise awareness amongst the public on the need to conserve water sources. You have been invited to attend address the public.

Task. As a chemistry student, prepare a short message you will use to deliver in the workshop. **End.**